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PROJECT REPORT

MCDA 5585 and MCDA 5586

Intelligent Automation and Reliability Engineering for Financial Services

Submitted By:

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under the guidance of:

Dr. Pawan Lingras

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in partial fulfillment for the awarding of the degree:

**Master of Science in
Computing and Data Analytics**

July 29, 2026

ACKNOWLEDGEMENT

I would like to express my heartfelt appreciation to everyone who contributed to the successful completion of this major project report. Their guidance, support, and encouragement were pivotal to its success.

I am especially grateful to **Dr. Pawan Lingras**, Academic Supervisor and Director of the Master of Science in Computing and Data Analytics (MCDA) program at Saint Mary's University, for providing me with this invaluable opportunity. His constant support and guidance were crucial throughout the duration of this project.

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Finally, I would like to thank both **Saint Mary's University** and **Citco Technology Management (CTM)** for giving me this incredible opportunity to gain industry exposure and work on such a sophisticated real-world financial technology project.

CERTIFICATE

This is to certify that **Bhavik Kantilal Bhagat** (Student ID: **A00494758**), a student at Saint Mary's University in the Master of Science in Computing and Data Analytics program, has completed the projects in fulfillment of the course requirements for **MCDA 5585** and **MCDA 5586** on the project titled “**Intelligent Automation and Reliability Engineering for Financial Services**”.

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ABSTRACT & EXECUTIVE SUMMARY

This report presents the design, technical implementation, data engineering pipelines, and business outcomes achieved during the major project internship at **Citco Technology Management Canada Limited** in partial fulfillment of the requirements for MCDA 5585 and MCDA 5586.

Statement of Hours:

This project represents a total of **900 hours of technical and project work** (March 15, 2026 – August 31, 2026) dedicated to fulfilling the degree requirements for the Master of Science in Computing and Data Analytics at Saint Mary's University.

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A Appendix: Hourly Activity Logs Summary

Below is the summary breakdown of technical and report writing hours for **MCDA 5585** and **MCDA 5586**:

Phase / Period	Key Tasks / Deliverables	Tech Hours	Writing Hours	Total
Phase 1 (Weeks 1–4)	Requirements, Pipeline Setup	50	5	55
Phase 2 (Weeks 5–8)	Architecture, Algorithm Implementation	50	5	55
Phase 3 (Weeks 9–12)	Fine-Tuning, Validation, Testing	50	10	60
Phase 4 (Weeks 13–16)	Deployment, PM & Report Writing	50	20	70
Total	Complete Deliverables Package	200	40	240

Table A.1: Summary of Technical and Report Writing Hours.

Note: The complete itemized weekly log timesheet is provided in the supplementary files `WorkLogs.xlsx` and `WorkLogs.pdf`.